# DISHA A

**Entry-Level Software Engineer | C, Linux, Computer Networking, Python**

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LinkedIn | GitHub

## PROFESSIONAL SUMMARY

BCA graduate with a strong foundation in C programming, computer networking (OSI/TCP-IP, L2 switching, L3 routing, OSPF and BGP fundamentals), Linux and data structures. Built two C projects on Linux: a raw-socket packet sniffer that parses Ethernet, IP, TCP, UDP and ICMP headers, and a Unix shell with pipes, redirection and signal handling. Comfortable with Git, GDB and debugging C programs on Linux. Eager to learn networking and system software as a Software Engineer on the IOS XE platform team.

## TECHNICAL SKILLS

**Programming:** C (pointers, structures, memory management, file handling), C++ (OOP, dynamic memory), Python (scripting and automation)
**Networking:** OSI and TCP/IP models, TCP/UDP, IP addressing and subnetting, Ethernet and ARP, L2 switching, L3 routing, OSPF and BGP fundamentals, ICMP
**Linux and System Programming:** Linux (Ubuntu/WSL), shell commands, processes (fork/exec), pipes, signals, sockets, Makefiles
**Debugging and Tools:** GDB, PDB, GCC, Makefiles, Git, GitHub
**Core CS:** Data Structures and Algorithms, Object-Oriented Design, DBMS
**Other:** Django, Flask, REST APIs, SQL/MySQL

## PROJECTS

**Packet Sniffer (C, Linux, Raw Sockets)** **GitHub Repository**

- Built a command-line packet sniffer in C using Linux AF_PACKET raw sockets that captures live traffic and parses Ethernet, ARP, IPv4, TCP, UDP and ICMP headers using structs and network byte-order conversion.
- Implemented filtering by protocol, IP address and port, optional hex dump, and a summary of per-protocol packet counts and top talkers on exit.
- Added bounds checks on every header before parsing, and tested against live traffic generated with ping and curl.

**Mini Unix Shell (C, Linux, Processes and IPC)** **GitHub Repository**

- Developed a Unix shell in C using fork, execvp and wait, supporting multi-stage pipes, input/output redirection (<, >, »), background jobs and built-in commands (cd, exit).
- Handled Ctrl+C with signal handlers so the foreground job stops while the shell keeps running, and reaped background processes to avoid zombies.
- Debugged with GDB and managed file descriptors carefully so pipes close correctly.

## EDUCATION

**Bachelor of Computer Applications (BCA)** 2023 – 2026
East West College of Management, Bengaluru CGPA: 7.85
**Pre-University Course (PUC)** 2020 – 2022
East West PU College, Bengaluru 60%
**Class X** 2020
Roan International School, Bengaluru 74%

## RELEVANT COURSEWORK

C Programming, C++ Programming, Computer Networks (OSI/TCP-IP, subnetting, L2 switching, L3 routing, OSPF, BGP), Data Structures and Algorithms

## LANGUAGES

English | Kannada | Hindi